

CRIS AI Block Scheduler

System Architecture, Operational Workflows & Comprehensive Page Specifications

Author: CRIS AI Systems Architecture · **Protocols:** COA-2.1, BDMS-3.0, FOIS-ICMS-2.0, MILP-XAI-1.0
Network Scope: 8,990 Stations · 5,208 Trains · 10 Winter Years (2016–2025 Dataset Calibration)
Mathematical Formulation: Mixed Integer Linear Programming (HiGHS Exact Solver) with Explainable AI (XAI)

1. Executive Summary & Problem Context

The **Centre for Railway Information Systems (CRIS) AI Block Scheduler** is a mission-critical platform engineered for Indian Railways (IR) to solve the complex mathematical trade-off between infrastructure maintenance (Track P-Way, Electrical OHE, and S&T; Signaling) and train punctuality. By unifying fragmented data streams across Control Office Application (COA), Block & Disruption Management System (BDMS), Freight Operations Information System (FOIS), and Integrated Coaching Management System (ICMS), the system automates multi-department **"Shadow Block"** synthesis and delivers mathematically optimal maintenance decisions via Mixed Integer Linear Programming (MILP).

2. Page Inventory & Operational Directory

#	Page Name	Route / URL	Core Operational Task & Engine	Protocol
1	Operations Command	/ (index.html)	Network-wide topology overview, protocol health aggregation, fleet punctuality tracking, and visual trend spectra.	System Hub
2	Dataset & Analytics	/analytics	10-Year historical delay exploration (2016–2025), multi-filter data inspection, YoY line analysis, and CSV dataset export.	Data Baseline
3	BDMS Shadow Block	/bdms	Multi-department engineering block co-registration (Track, OHE, Signaling) synthesizing single unified disruption windows.	BDMS-3.0
4	MILP Optimizer Studio	/milp	HiGHS MIP solver evaluating binary decisions ($x_j \in \{0,1\}$), parameter sensitivity tuning (λ), SHAP attribution, and XAI audits.	MILP-XAI-1.0
5	COA Live Telemetry	/coa	100ms real-time locomotive telemetry, speedometer, traction current, KAVACH ATP link, line berth occupancy, and station logs.	COA-2.1
6	FOIS Priority Matrix	/fois	Dynamic priority formula ($P_i \in [1,10]$) decomposition, cascading delay budgets (Δd_i), and financial SLA revenue risk modeling.	FOIS-ICMS-2.0
7	Operations Terminal	/terminal	Interactive CLI console for dispatchers: status health checks, batch corridor optimization, fleet dumps, and latency diagnostics.	CLI Dispatch

3. Detailed Technical Breakdown by Page

Page 1: Operations Command & CRIS Network Topology (index.html)

Core Purpose: Serves as the executive gateway and overall network health dashboard, displaying live IR network status, summary KPIs across all 4 protocols, and comprehensive fleet monitoring.

Key Tasks & Live Functions:

- **Network Scope Telemetry:** Displays 8,990 Stations, 5,208 Trains, 10-Year historical delay baseline, and HiGHS MIP optimality status.
- **Flagship Fleet Monitoring Table:** Interactive table rendering 10 flagship train services enriched with Zone, Route, Stops, 10-Yr Mean Delay, Dynamic Priority (P_i), Allowed Delay Threshold (Δd_i), Sensitivity band, and Punctuality percentage.
- **10-Year Historical Mean Delay Chart:** Bar visualizer illustrating average winter operational delays per service (ranging from 19.8m to 62.3m).
- **Priority vs. Delay Threshold Bubble Chart:** Dynamic 2D spectrum plotting train delay tolerances against priority weights.

Page 2: Real Dataset & Historical Delay Analytics (pages/analytics.html)

Core Purpose: Deep statistical analytics studio for the 100 historical winter delay records across 10 flagship trains (2016–2025).

Key Tasks & Live Functions:

- **100-Record Observation Explorer:** Complete historical log displaying Train ID, Name, Number, Route, Year, Scheduled Arrival, Actual Arrival, Delay Minutes, and Punctuality Classification.
- **Live Multi-Parameter Filtering:** Real-time search bar and year selector (All Years or individual years 2016–2025) with instant record count updates.
- **Year-over-Year Trajectory Line Chart:** 10-year comparative line chart tracking punctuality fluctuations over 10 winter seasons.
- **CSV Dataset Export:** One-click generation and browser download of the complete 100-row standard CSV dataset.

Page 3: BDMS & Shadow Block Synchronization Engine (pages/bdms.html)

Core Purpose: Protocol 2 engine that merges multi-department maintenance requests into unified Shadow Blocks, preventing redundant corridor shutdowns.

Key Tasks & Live Functions:

- **Corridor Candidate Selection:** Interactive dropdown for 4 real IR corridors (Northern Trunk, Central Grand Trunk, Southern Ocean Trunk, Eastern Coal Corridor).
- **Multi-Department Co-Registration:** Synchronizes Track (P-Way), Electrical (OHE), and Signaling (S&T;) into a single non-conflicting time window.
- **Efficiency Gain Calculation:** Computes saved train-minutes (e.g. 840 train-minutes saved by merging 390 total requested minutes into a 165m window).
- **Affected Fleet Impact Table:** Identifies trains operating through the section and calculates their priority and delay budget.
- **Live BDMS JSON Stream:** Generates validated `cris.bdms.block-request.v3` payloads with one-click clipboard copying.

Page 4: MILP Optimizer & Explainable AI Studio (pages/milp.html)

Core Purpose: Mathematical optimization and Explainable AI (XAI) transparent audit workbench.

Key Tasks & Live Functions:

- **Exact HiGHS Solver Execution:** Evaluates maintenance yield M_j against total disruption penalty $\lambda \cdot \sum(P_i \cdot \Delta d_i)$ to produce binary decision x_j in $\{0, 1\}$.

- **Interactive Sensitivity Workbench:** Sliders for Penalty Scaling (λ : 0.1 to 5.0) and Traffic Window Offset (+0h to +4h) with real-time recalculation of decision and net objective value.
- **Per-Train Disruption Impact Table:** Granular table showing individual train delay minutes, SLA financial exposure (INR), and penalty contributions.
- **SHAP Feature Attribution:** Transparent natural language and metric breakdown explaining why the decision was approved or rejected.
- **Counterfactual 'What-If' Scenarios:** Automatic evaluation of alternative maintenance timing windows.
- **Live MILP JSON Stream:** Outputs `cris.milp.optimizer-output.v1` schema.

Page 5: COA Live Telemetry & Ingestion Lab (pages/coa.html)

Core Purpose: Protocol 1 high-frequency telemetry ingestion lab simulating real-time train describer data.

Key Tasks & Live Functions:

- **Locomotive Cockpit Gauges:** Real-time animated speedometer (km/h) with live jitter, Traction Current (A), Brake Pressure (5.0 bar), ATP status, and GSM-R link.
- **Route Topology & Section Berths:** Real-time track progression bar and section line berth matrix (Main UP: Occupied, Main DOWN: Clear, Loop Line: Reserved).
- **Station Event Log & Punctuality Tracker:** Live station log tracking scheduled vs. actual arrival/departure and crew changes.
- **Live COA JSON Stream:** Emits `cris.coa.train-telemetry.v2` payloads at 100ms ingestion intervals.

Page 6: FOIS & ICMS Dynamic Priority Intelligence (pages/fois.html)

Core Purpose: Protocol 3 intelligence module computing dynamic priority weights and cascading delay budgets.

Key Tasks & Live Functions:

- **Dynamic Priority Decomposition:** Computes $P_i = \alpha * B_{cat} + \beta * L_{dist} + \gamma * V_{vol} + \delta * C_{casc}$ ($\alpha + \beta + \gamma + \delta = 1.0$).
- **Delay Budget (Delta d_i) Allocation:** Calculates allowable tolerance limits before exponential penalty scaling is incurred.
- **Financial SLA Risk Modeling:** Computes passenger load, PNR revenue (INR Lakhs), and SLA penalty breach rates (up to Rs 15,000/min).
- **10-Train Priority Spectrum Chart:** Dual-axis bar chart comparing P_i and Delta d_i across the entire fleet.

Page 7: CRIS Central CLI & Operations Terminal (pages/terminal.html)

Core Purpose: Low-latency operational terminal and command-line dispatch console for system controllers.

Key Tasks & Live Functions:

- **Interactive CLI Shell:** Supports commands: status (protocol diagnostics), solve (batch MILP solver), fleet (dump priority matrix), telemetry (force GPS refresh), ping (server latency), and clear.
- **One-Click Quick Actions:** GUI buttons to execute instant operational dispatch routines.

4. Architectural Standards & Design Enforcement

- **Strict 100% Crisp White Typography:** All chart bars, bubble points, axis labels, legend titles, and table cells utilize #ffffff and #f1f5f9 for maximum legibility over dark glassmorphic backdrops.
- **Clean Plain Pink Branding:** Navigational active states and badges are rendered in flat plain pink (#ec4899) without glowing neon blooming.
- **Original Full-Color High-Resolution Train Photography:** Page backdrops utilize full-saturation train photographs (High-speed HST, WDP4D diesel locomotive, Vande Bharat, etc.) with dark obsidian glass overlays (rgba(10, 13, 22, 0.88)).
- **Cache-Busting Netlify Deployment:** All CSS assets and background images enforce versioned query parameters (?v=3) for instant CDN refresh upon deployment.